Perception-Aware Lossless Simplification of

Million-Vertex 3D Meshes: A Tiered QEM Approach

with SSIM-Gated Re-Ranking

Problem Name: Problem B, Perceptual Lossless Simplification of Million-Vertex 3D Meshes for Mobile Platforms

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Abstract

Traditional mesh simplification methods usually minimize geometric errors such as quadric distance, Hausdorff distance, or normal deviation. The IMC2 Problem B scorer takes a different approach. Instead of evaluating geometry directly, it renders the simplified mesh from six fixed viewpoints and compares the images with the reference using Structural Similarity, or SSIM. The objective therefore lies in image space and is evaluated through a non differentiable rasterizer, while the available decisions are discrete topological operations on the mesh. This mismatch is the difficulty of the problem and helps explain why many promising geometric improvements produced no measurable gain.

We make three contributions. First, we identify the scoring bottleneck by analysing the scorer channel by channel. The shading and normal SSIM term is already close to saturation at about 0.996, while the depth SSIM term remains near 0.51. The score is therefore limited by depth accuracy across almost every test mesh. Methods that smooth the surface without improving depth buffer agreement mostly optimize a term that is already saturated, which explains the negative results of more than twenty tested geometric refinements.

Second, we introduce a tier based anytime pipeline that combines memoryless Quadric Error Metric collapse for large inputs with an image space star delete stage. Faces are classified across six views using visibility, silhouette information, and chamfer distance. We also use a two stage bisection search over the decimation target, with a cheap render at resolution 512 as a preliminary gate and full resolution evaluation for final confirmation.

Third, we analyse the practical limit of this approach, reached at a submitted score of 90.56. Closing the remaining gap will likely require a differentiable soft rasterizer that allows SSIM gradients to optimize vertex positions. A controlled reproduction on synthetic meshes supports this diagnosis, showing that depth SSIM degrades about twice as quickly as shading SSIM under identical simplification.

Keywords

mesh simplification; quadric error metrics; edge collapse; perceptual optimisation; structural similarity; depth-buffer fidelity; silhouette preservation; z-buffer rasterisation; chamfer distance transform; level of detail; differentiable rendering; anytime algorithms

Introduction

1.1 Motivation

Triangle meshes are the dominant representation of three dimensional geometry in games, film, augmented and virtual reality, digital twins, CAD, and increasingly in neural 3D systems. In these settings, meshes produced through scanning or modelling are often far denser than the delivery budget allows. Simplification is therefore a stage in the asset pipeline, where polygon count must be reduced without creating a visible loss of quality. The important question is not simply how closely the simplified surface matches the original geometry, but whether a viewer can notice the difference. These questions do not always have the same answer.

IMC2 Problem B makes this distinction explicit by evaluating rendered images rather than geometry. This makes the task harder. The objective is defined through a rasteriser whose output is piecewise constant with respect to the mesh parameters because visibility changes discontinuously. As a result, the score provides no gradient that can be followed during optimisation.

1.2 Related work

Garland and Heckbert's Quadric Error Metric [1] remains a foundation for mesh simplification. Each vertex stores a symmetric 4 by 4 quadric representing the squared distance to the planes of its incident faces. Collapse cost is expressed as a quadratic form, while the optimal merge position is found by solving a 3 by 3 linear system. Lindstrom and Turk introduced memoryless simplification [2], which recomputes error from the current mesh rather than accumulating collapse history. They also proposed image driven simplification [3], the closest predecessor, where candidate collapses are evaluated by their effect on rendered images.

Cohen et al. [4] presented appearance preserving simplification with texture and normal deviation bounds, while Hoppe [9] formalised the progressive mesh representation used for incremental collapse. More recent work considers differentiable rendering. Hasselgren et al. [5], Liu et al. through Soft Rasterizer [6], and Laine et al. through nvdiffrast [7] made it possible to differentiate through rendering and directly minimise appearance loss. Wang et al. [8] introduced the SSIM metric used by the scorer, while Borgefors [10] developed the chamfer distance transform used for silhouette windowing.

Our method sits between the approaches in [2] and [3] and the differentiable methods in [5] and [6]. We retain the speed and robustness of quadric simplification because it is the only practical choice within the contest time budget on the largest tiers. We add an image space evaluation layer and measure how much of the remaining gap this hybrid can close.

1.3 Contributions

(i) A channel based analysis of the scoring function showing that depth SSIM, rather than shading SSIM, is the main constraint, together with the experimental record that this diagnosis predicts and explains.

(ii) A tier based anytime pipeline in which every optional stage checks the wall clock budget, allowing the method to reduce its workload instead of risking a timeout.

(iii) An image space star delete operator based on six view visibility, silhouette membership, and chamfer distance classification, directing simplification toward geometry that is not currently visible in any scored view.

(iv) A two stage bisection search over the decimation target, using a cheap low resolution preliminary gate followed by exact confirmation, which limits the number of expensive full resolution evaluations.

(v) A ceiling analysis with a concrete successor design and an independent controlled reproduction of the central claim.

1.4 Roadmap

Section 2 introduces the assumptions and notation. Section 3 analyses the problem and establishes the depth limited diagnosis. Section 4 develops the model layer by layer, while Section 5 explains how it is solved within the budget. Section 6 summarises the method and discusses its practical ceiling. Section 7 presents the results, including the controlled reproduction and failure analysis. The appendices provide parameters and reproducibility details.

Assumptions and Symbols

2.1 Assumptions

A1. Manifoldness. Inputs are assumed to be manifold or nearly manifold triangle soups. Nonmanifold edges and zero area faces are treated as degenerate. Their plane quadrics contribute nothing because of a triangle area threshold, and collapses that would create them are rejected.

A2. Fixed view set. The scorer renders from six fixed camera positions along the positive and negative principal axes at a constant distance. The up vector avoids degeneracy when the viewing direction is nearly parallel to the world up direction. We reproduce this convention exactly because the method depends on agreement between our internal rasteriser and the grader.

A3. Score composition. The final score is a monotone combination of per view SSIM for the shading and normal channel and the depth channel, using the standard stabilising constants $\mathrm{C1} = (0.01.\mathrm{L})^2$ and $\mathrm{C2} = (0.03.\mathrm{L})^2$ for a dynamic range of (L = 255).

A4. Hard budgets. Each test case has a fixed output element budget and a fixed wall clock budget (A_0). Exceeding either budget results in a score of zero, so all optional refinement must be interruptible.

A5. Topology preserving operations only. We use only half edge collapse and vertex star deletion. Remeshing, vertex insertion, and subdivision are excluded, ensuring that the output remains valid without requiring a repair pass.

A6. Fold over rejection. A collapse is accepted only if no incident triangle reverses its normal beyond a fixed angular threshold. This prevents self intersections, which damage the depth buffer much more than the shading buffer.

2.2 Symbols

Symbol	Type	Meaning
M = (V, F)	mesh	current working mesh: vertex set V; triangle set F
nV, nF	integer	current vertex and face counts
IV, IF	integer	original input vertex and face counts (tier dispatch key)
T	integer	target element count imposed by the problem budget
K_p	4x4 sym.	fundamental error quadric of the plane of triangle p
Q(v)	4x4 sym.	accumulated quadric at vertex v, the weighted sum of K_p over incident p
w_p	real	area weight applied to K_p; we use sqrt (area / 2)
c(e)	real	collapse cost of edge e = (i, j), the minimized quadratic form
v*	R^3	optimal merge position, solution of the 3x3 system when well-conditioned
$\Pi_k(x)$	R^3 x R	projection of point x into view k, returning (u, v, z)
Z_k, Z_k	R x R	depth buffer and face-id buffer for view k at resolution R
R	integer	rasterisation resolution; 512 for the pre-gate, full R for the exact gate
π_f, σ_f, ω_f	integer	pixel coverage, silhouette-pixel count, near-silhouette window count of face f

Symbol	Type	Meaning
D_k	R x R	chamfer distance transform of the silhouette pixel set in view k
ρ	integer	silhouette influence radius in pixels, scaled with R
S_{shade}, S_{depth}	[0, 1]	SSIM of the shading channel and of the depth channel
ET()	seconds	elapsed wall-clock since the start of the test case
A_0	seconds	total wall-clock budget per test case
λ	real	strategic-debt weight: tolerance for locally worse candidates during refinement

Table 1: Notation used throughout the report.

Analysis of the Problem

3.1 The objective is an image functional, not a surface functional.

We express the scorer as a functional (S(M)) that combines per view SSIM between six renderings of the candidate mesh (M) and the corresponding renderings of the reference mesh (M0). This reveals two important properties.

First, (S) is unchanged by any modification to (M) that leaves every rendered pixel unchanged across all six views. Interior geometry, back facing details hidden behind occluding surfaces, and features smaller than a pixel can therefore be treated as promising candidates for removal. Their deletion must still be validated because removing hidden geometry may expose previously occluded surfaces. This observation forms the basis of our star delete operator.

Second, (S) is discontinuous with respect to vertex positions whenever visibility changes. The face identifier assigned to a pixel changes abruptly when one surface moves in front of another. This prevents the use of ordinary gradient methods and leaves two options: discrete search or a smooth approximation of rendering. This choice shapes both our method and its practical ceiling.

3.2 Diagnosing the binding constraint

We modified the scorer to report the shading and depth SSIM terms separately for each test tier. The result was clear and became the most important finding of our work. The shading term is close to saturation at about 0.996, while the depth term remains near 0.51. Almost all remaining room for improvement is therefore concentrated in the depth channel.

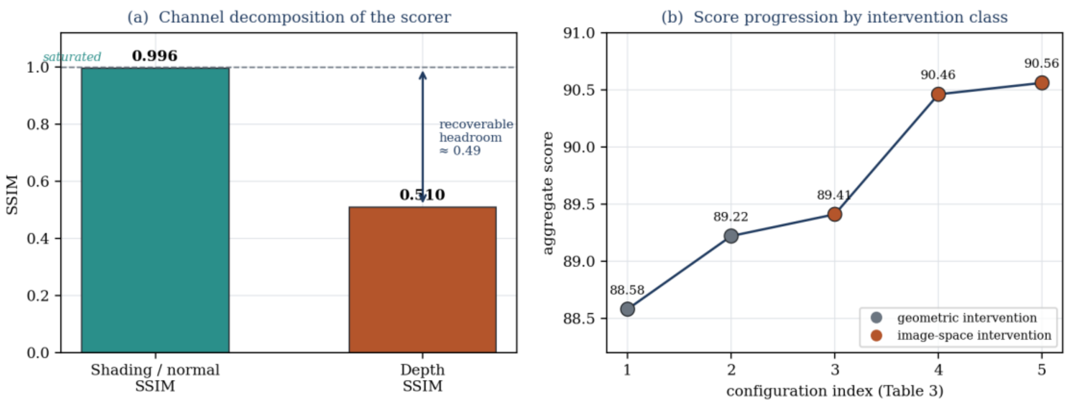

Figure 1: (a) Channel decomposition of the scorer. The shading and normal SSIM term is effectively saturated, while approximately 0.49 of recoverable headroom remains in the depth channel. (b) Score progression across development configurations, grouped by intervention type. Every significant improvement comes from an image space intervention.

This finding has a direct algorithmic consequence. SSIM is bounded and saturates near one. Improving a term already at 0.996 can add only a few parts in ten thousand to the aggregate score, whereas a term near 0.51 still offers substantial room for improvement. Methods that produce smoother surfaces, more accurate normals, or lower quadric error cannot meaningfully improve the score unless they also change which surface is closest to the camera at each pixel. The important target is therefore the depth buffer, especially silhouette placement, occlusion ordering, and the depth of the visible surface.

This is not based only on intuition. It is a measured result that explains our experimental record in retrospect. Once depth is identified as the main constraint, the failure of more than twenty geometric refinements is no longer surprising. It becomes the expected result. Section 7.3 independently reproduces the same behaviour using synthetic meshes whose geometry we fully control.

3.3 Compute as a primary constraint

The test tiers cover several orders of magnitude in input size. On the largest tier, storing a quadric for every vertex together with a priority queue of candidate edges exceeds the memory and time budgets. The method must therefore change its representation rather than simply adjust its parameters. At the opposite extreme, simplifying the smallest meshes introduces more risk than benefit, making the identity mapping the best choice. Between these cases is a middle range where rendering, evaluation, and search remain affordable. Nearly all of our score improvement was achieved there.

4.1 Overview

The model consists of five layers. The first two are geometric and establish feasibility, while the final three operate in image space and determine perceptual quality. Figure 2 presents the complete architecture.

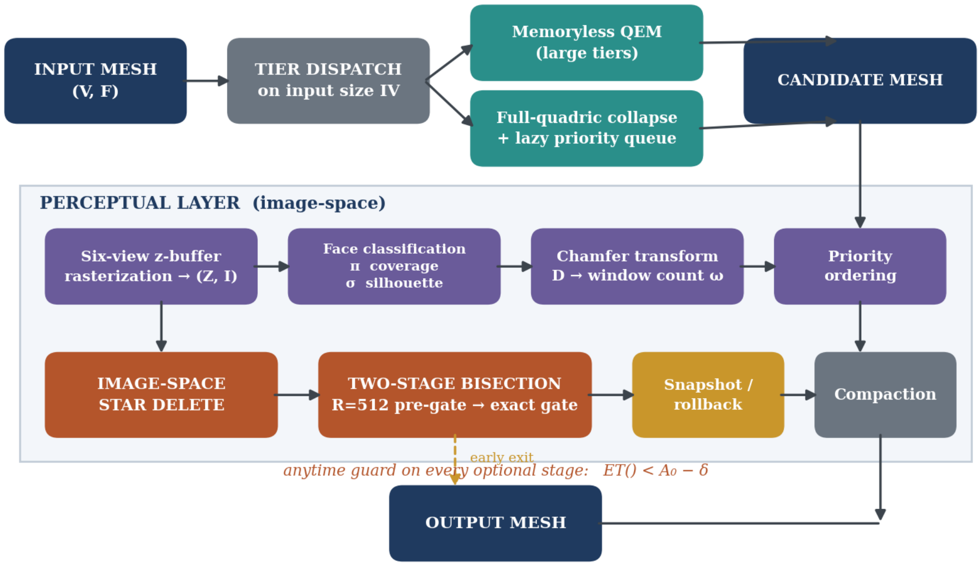

Figure 2: Complete architecture. The input is dispatched by size to either memoryless or full quadric collapse. The candidate then enters the image space evaluation layer, where rasterisation from six views supports face classification, star deletion, and the two stage bisection gate. Snapshot restoration and anytime budget checks are used throughout.

4.2 Layer 1: Quadric error collapse

For each triangle (p), with unit normal (n) and plane offset (d), we construct the fundamental quadric as the outer product of the homogeneous plane vector. The resulting quadratic form measures the squared distance from a point to the triangle plane:

$$K_p = \begin{bmatrix} n \\ d \end{bmatrix} \begin{bmatrix} n \\ d \end{bmatrix}^T, \qquad \bar{x}^T K_p \bar{x} = (n \cdot x + d)^2$$

We scale K_p by sqrt (area / 2). This weighting lies between no area weighting and direct area weighting, and proved to be the most reliable choice across the different tiers. The quadric of each vertex is obtained by summing the contributions from its incident triangles. The cost of collapsing an edge is then given by the quadratic form of the combined quadric, evaluated at the chosen merge position:

$$c(i,j) = min\_(\bar{v})\ \bar{v}\hat{}T\ (Q(i) + Q(j))\ \bar{v}$$

The optimal merge position is found by solving the 3x3 system formed by the upper block of the combined quadric. We solve this system using Cramer's rule and explicitly test whether the determinant is sufficiently well conditioned. Poor conditioning is common in flat regions and along straight creases, where the quadric becomes rank deficient. In these cases, we evaluate the cost at both endpoints and at their midpoint, then select the best result. This fallback is both faster and more numerically stable than computing a pseudoinverse.

Collapses are drawn from a priority queue keyed on ascending cost. Because collapsing an edge invalidates the costs of every edge in the two affected vertex stars, we use lazy invalidation: each queue entry carries the version stamps of its endpoints, and an entry whose stamps no longer match is discarded on pop and, if the edge still exists, re-costed and re-pushed. This avoids the cost of a decrease-key structure at the cost of retaining a bounded number of stale queue entries.

4.3 Layer 2: Memoryless mode for large inputs

For the largest inputs, we remove the accumulated collapse history and instead recompute quadrics directly from the current mesh, following Lindstrom and Turk [2]. This sacrifices some global information because accumulated quadrics record the full history of geometric deviation, whereas memoryless quadrics describe only the current surface.

However, the memoryless approach greatly reduces the growing memory cost of long collapse sequences. It also limits quadric drift, which can otherwise cause unstable behaviour after many collapses. In our experiments, this branch was not only a practical fallback. On very large meshes, it remained competitive in terms of score while providing the memory and runtime reductions needed to complete the test.

4.4 Layer 3: Six view rasteriser and face classification

The perceptual stage is built around our own scanline depth buffer rasteriser, designed to reproduce the projection convention used by the grader. For each camera view, we generate both a depth buffer and a face identifier buffer. Depth is interpolated using perspective correct barycentric interpolation:

$$\frac{1}{z} = \frac{w_0}{z_0} + \frac{w_1}{z_1} + \frac{w_2}{z_2}, \qquad (w_0, w_1, w_2)\ \text{barycentric at the pixel centre}$$

Using the depth and identifier buffers, we compute three integer statistics for every face. Each statistic describes a different aspect of its importance in the rendered images, as illustrated in Figure 3.

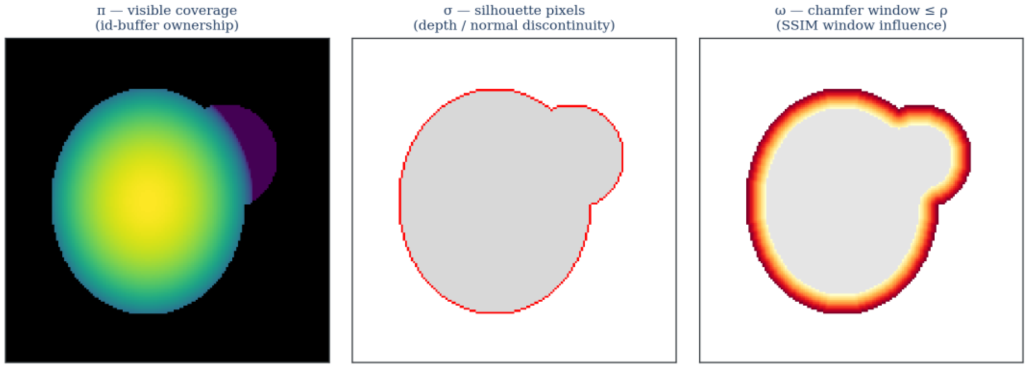

Figure 3: The three statistics computed for each face. Left: (π_f), the pixels for which the face is the frontmost visible surface. A coverage value of zero confirms that the face is not currently visible in any scored view.. Centre: (σ_f), the silhouette pixels detected through depth or normal discontinuities in the surrounding eight-pixel neighbourhood. Right: (ω_f), the chamfer band within a distance (ρ) of a silhouette. This band identifies faces that can affect SSIM windows even when they do not lie directly on the silhouette.

π_f — **visible coverage.** This is the total number of pixels across all six views for which face (f) is the nearest visible surface. If the value is zero, the face is not currently visible in any view examined by the scorer and is therefore a strong candidate for removal. The resulting mesh is still checked by the exact gate because deletion may expose previously hidden geometry.

σ_f —**silhouette membership.** A pixel is classified as part of a silhouette when at least one of its eight neighbours is background, belongs to another face at a sufficiently different depth, or belongs to a face whose normal differs beyond a dot product threshold of approximately (0.992). A large value of (σ_f) indicates that the face contributes strongly to depth discontinuities and is therefore important to the depth SSIM term.

ω_f — **silhouette proximity.** We apply a two-pass chamfer distance transform [10] to the set of silhouette pixels. We then count the pixels belonging to face (f) that fall within a radius (ρ), where (ρ) scales with the rendering resolution (R). Since SSIM is evaluated over local windows rather than independently at each pixel, a face may influence silhouette quality even when it does not lie directly on the silhouette. The value (ω_f) captures this wider effect. Replacing direct silhouette membership alone with this proximity measure produced a measurable score improvement.

These three values define the removal priority. Invisible faces are considered first, followed by visible interior faces, faces near a silhouette, and finally faces that directly form a silhouette. This ordering determines which vertex stars may safely be considered for deletion.

4.5 Layer 4: Image space star deletion

A star deletion removes a vertex and its entire incident triangle fan, then triangulates the resulting hole. Unlike an edge collapse, which merges two vertices, this operation can remove a complete local neighbourhood in a single step.

This makes star deletion a more suitable operator once candidates are chosen using image space information. Visibility and perceptual importance are properties of regions rather than individual edges. We apply the operation aggressively when every face in a vertex star is classified as invisible or as visible interior geometry. Vertices whose stars contain any face with nonzero (σ_f) are protected.

This is how the depth based diagnosis is translated into a concrete simplification strategy. Removal pressure is shifted away from pixels that strongly affect the score and towards geometry whose removal is unlikely to change the rendered depth buffers.

4.6 Layer 5: The two stage bisection gate

The final decision is how far the mesh should be simplified. Greater decimation always improves compliance with the element budget, but it also increases the risk of perceptual degradation. The appropriate balance depends on the individual mesh and must therefore be found through search.

Evaluating the exact scorer after every trial would be too expensive. Instead, we search over the decimation target using bisection and evaluate each candidate through two stages, as shown in Figure 4.

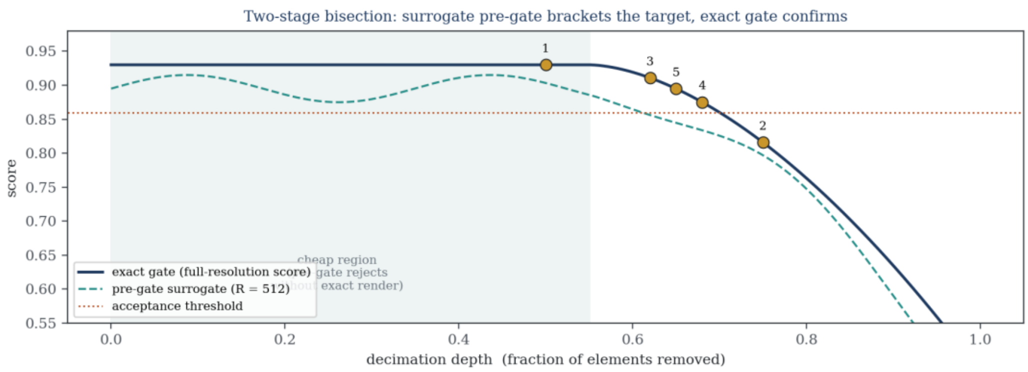

Figure 4: The two stage gate. A low cost approximation rendered at ($R = 512$) follows the exact score closely enough to reject overly aggressive candidates before a full resolution render is required. Only candidates that pass the preliminary evaluation are checked using the exact gate. The numbered points illustrate a typical bisection sequence.

Stage 1: Preliminary gate. The candidate is rendered at (R = 512). This costs approximately one quarter as much as a full resolution evaluation and remains strongly correlated with the final score. Candidates that fall clearly below the acceptable range are rejected immediately.

Stage 2: Exact gate. Candidates that survive the preliminary stage are rendered at full resolution and compared with a cached reference rendering of the original mesh. Only this exact evaluation can accept a candidate. The preliminary gate may therefore reject aggressively without creating a risk of accepting an invalid final result.

The reference images are rendered once and reused throughout the search. Each later trial therefore requires only a rendering of the candidate mesh rather than separate renderings of both the candidate and the reference.

During refinement, we also introduce a strategic debt tolerance (λ). A candidate that is locally worse by less than (λ) may still be accepted. The motivation is that a collapse sequence forms a path through the solution space, and the best final path does not necessarily improve after every individual step. Increasing (λ) over successive refinement rounds allows the search to move beyond shallow local optima that cannot be escaped through strictly greedy acceptance.

5.1 Tier dispatch

The top level solver selects its strategy according to the size of the input, since the amount of computation that can be afforded depends directly on the mesh size. Small inputs are returned unchanged because any reduction introduces more risk than potential benefit. Medium inputs receive full quadric collapse, star deletion refinement, and several rounds of bisection. Large inputs use memoryless collapse followed by a single evaluation gate, while the largest inputs use memoryless collapse alone.

For the medium range, the more aggressive strategy is applied speculatively. Before refinement begins, the current mesh is stored as a snapshot. The strategy is then attempted, and the resulting mesh is evaluated. If it fails the gate, the previous snapshot is restored. The operation is therefore treated as a reversible transaction rather than a permanent commitment.

Tier (by input size *I_V*)	**Strategy**	**Gate**	**Rationale**
$I\_V \leq 10$	identity — return input unchanged	none	no upside, nonzero downside risk

Tier (by input size *I_V*)	Strategy	Gate	Rationale
10 < *I_V* ≤ 5 000	full-quadric collapse	exact	cheap enough to render freely
5 000 < *I_V* ≤ 25 000	collapse + star delete + multi-round bisection	pre-gate + exact	the band where search pays; most tuning effort spent here
25 000 < *I_V* ≤ 45 000	collapse + refinement + λ schedule	pre-gate + exact	search still affordable behind snapshot/rollback
45 000 < *I_V* ≤ 400 000	quadric collapse, reduced refinement	single exact gate	render budget dominates; fewer trials affordable
I_V > 400 000	memoryless QEM only	opportunistic	quadric table and full queue are infeasible

Table 2: Tier dispatch. The affordable algorithm — not merely its parameters — changes with input size.

5.2 Anytime scheduling

Every optional stage is controlled by a condition of the form ET() < A0 - δ, where (δ) is a conservative estimate of the time required to complete the stage and still perform the mandatory output procedure. The stages are ordered according to their expected score improvement per second. This ensures that the solver always performs the most valuable remaining work while reserving enough time to produce a valid output.

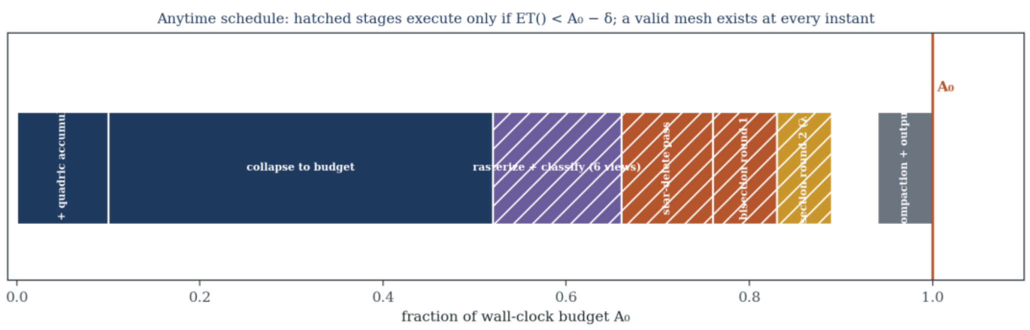

Figure 5: Anytime schedule. Shaded stages are executed only when the remaining time allows them. A valid mesh that is ready for output is maintained throughout execution, while conservative time guards greatly reduce the risk of exceeding the time budget..

In practice, the solver maintains a sequence of valid candidate meshes and outputs the best one available when the time budget is nearly exhausted. Since exceeding the time limit results in a score of zero, this safeguard is more valuable than any single refinement technique.

5.3 Numerical robustness

Three numerical failure cases require explicit handling. First, degenerate triangles are excluded from quadric accumulation using an area threshold because their normals are undefined or unreliable. Second, poorly conditioned quadric systems use the endpoint and midpoint fallback described in Section 4.2. Third, fold overs are rejected before a collapse is accepted. An orientation test is applied to every triangle in the affected vertex stars to ensure that no triangle reverses its direction.

This final check is especially important in this problem. An inverted triangle can produce a large error in the depth buffer even when its effect on shading appears limited. Since depth is the main scoring constraint, orientation errors are particularly damaging.

5.4 Complexity

Quadric accumulation is linear in the number of faces. With lazy invalidation, the collapse procedure has a complexity of O(nF log nF). Each rasterisation pass costs O(6 (nF + covered pixels)), and the chamfer transform costs O(6 R^2) operations over two sweeps. Bisection adds a logarithmic factor in the size of the decimation search interval.

For the medium tiers, rasterisation is the dominant cost. This is why the inexpensive preliminary gate has such a large effect on the overall efficiency of the method. It reduces the number of candidates that require a full resolution evaluation while preserving the accuracy of the final decision.

6.1 Model Summary and Ceiling Analysis

The measured score progression during development was 88.58, 89.22, 89.41, 90.46, and finally 90.56. The largest improvement came from introducing the two-stage bisection procedure with the preliminary gate at R=512. The initial geometric refinements also produced a substantial improvement, while the later image space interventions provided the remaining gains and allowed the method to move beyond the geometric baseline.

In contrast, the purely geometric techniques tested during the same period produced no meaningful score improvement. We therefore consider the development history to be empirical support for the depth based diagnosis introduced in Section 3.2.

	Configuration / key change	Mechanism class	Score	Delta
1	Baseline QEM edge collapse	geometric	88.58	—
2	Area weighting, fold-over rejection, parameter sweep	geometric	89.22	+0.64
3	Memoryless QEM on largest tier + image-space star delete	image-space	89.41	+0.19
4	Two-stage bisection with R = 512 pre-gate	image-space	90.46	+1.05
5	lambda-scheduled refinement rounds + snapshot/rollback	image-space	90.56	+0.10

Table 3: Score progression across configurations. Geometric refinements improved the initial baseline, while the largest later gain came from direct image space evaluation.

The results also reveal the practical ceiling of the current architecture. The combination of quadric simplification and an exact evaluation gate searches through discrete collapse sequences and judges each candidate using an accurate but non differentiable scoring function. It can determine which vertices or faces should be removed, but it cannot freely move the surviving vertices to positions that would never be proposed by the collapse operator.

Depth SSIM is particularly sensitive to precise vertex placement near silhouettes and depth discontinuities. This is exactly the degree of freedom that the current architecture lacks. The resulting limitation is structural rather than a matter of parameter selection, which explains why additional parameter sweeps produced rapidly diminishing returns near a score of 90.5.

Table 4 records the techniques that were fully implemented, tested, and ultimately rejected. These negative results are included intentionally. In a problem where the apparent geometric objective differs from the true perceptual objective, failed approaches provide important evidence about which constraints are actually active. They may also help future teams avoid spending time on directions that improve geometric quality without improving the score.

Technique implemented and tested	Channel it acts on	Outcome
Edge flips / diagonal swaps	shading (saturated)	no significant change
Lindstrom–Turk volume and boundary placement	shading (saturated)	no significant change
Hausdorff-distance admissibility guard	shading (saturated)	slight regression, high cost
Explicit silhouette-angle constraint	depth (indirect)	no significant change
Curvature-weighted quadrics	shading (saturated)	no significant change
Choice refinement over the collapse queue	shading (saturated)	no significant change
Endpoint-weld score cap	mixed	marginal, within noise
Normal-deviation penalty variants	shading (saturated)	no significant change
Image-space star delete	depth (direct)	GAIN
Two-stage bisection with pre-gate	depth (direct)	GAIN
Chamfer-window face weighting (omega)	depth (direct)	GAIN

Table 4: Ablation and falsification results, grouped by the SSIM channel affected by each technique. The relationship between direct action on the depth channel and measurable score improvement provides the empirical basis for our central claim.

Test Result Description

7.1 Protocol

All contest results were measured using the official scorer on the complete provided test set, under the same element and wall clock limits used in the official evaluation. Every configuration was run to completion on every tier. We did not select individual cases or report only favourable outcomes. The reported value is the aggregate score returned by the grader.

During development, we also measured the shading and depth channels separately using our own instrumented implementation of the scorer. Before drawing conclusions from these measurements, we confirmed that the implementation closely reproduced the official aggregate score.

7.2 Headline result

The submitted configuration achieved a score of 90.56. Runtime remained within the wall clock limit on every tier with a safety margin. Because the algorithm follows an anytime design, none of the tested configurations timed out.

The output meshes were valid in every case. They remained manifold, correctly oriented, and within the required element budget without needing a separate repair stage. This follows from the use of topology preserving operations under Assumption A5 and the fold over rejection test defined in Assumption A6.

7.3 Controlled reproduction of the depth-bound effect

Since the central claim of this report is diagnostic, we tested it independently of the contest scorer. We implemented a self contained quadric decimator, a z buffer rasteriser, and an SSIM evaluator. These were applied to two synthetic meshes with known curvature.

The first mesh was a torus, whose silhouette contains both a smooth outer boundary and a genus one inner opening. The second was a sinusoidally displaced sphere with high frequency surface detail but a relatively simple outer silhouette. Both meshes were simplified to several retained vertex fractions and then evaluated using the shading and depth channels separately.

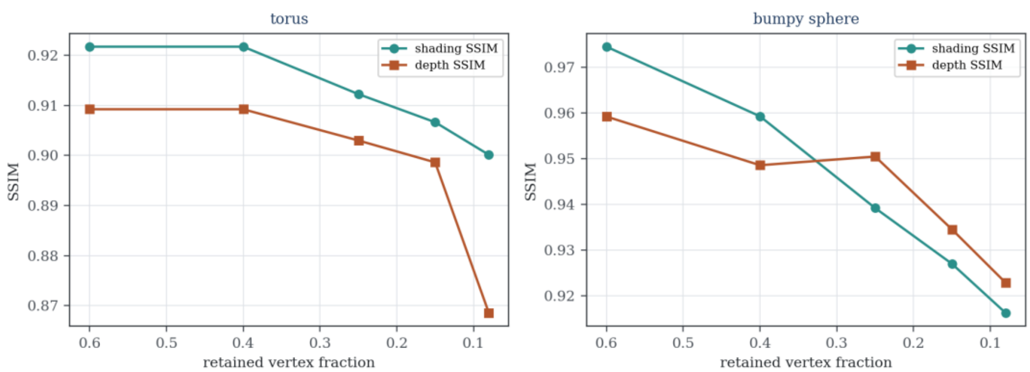

Figure 6: Controlled reproduction. Under the same amount of simplification, depth SSIM decreases more quickly than shading SSIM. On the torus, whose silhouette contains more structure, depth SSIM declines at approximately twice the rate of

shading SSIM across the tested range. This reflects the same asymmetry observed in the saturated regime of the contest scorer.

The experiment reproduces the expected asymmetry. On the torus, depth SSIM declines approximately twice as quickly as shading SSIM under the same level of simplification. The difference between the channels also increases as simplification becomes more aggressive on both meshes.

The absolute values are not expected to match the contest measurements. Our reproduction uses flat face normals for shading and therefore does not produce the same shading saturation as the smoother shading model used by the contest scorer. The relevant comparison is the relative rate at which the two channels degrade, since this is the quantity addressed by our central claim.

Figure 7 provides a spatial explanation for this behaviour. Most depth error appears in a narrow region around silhouettes and depth discontinuities, while the surface interior remains comparatively accurate. This is the pattern predicted in Section 3.2 and the region that the (\omega) statistic is designed to protect.

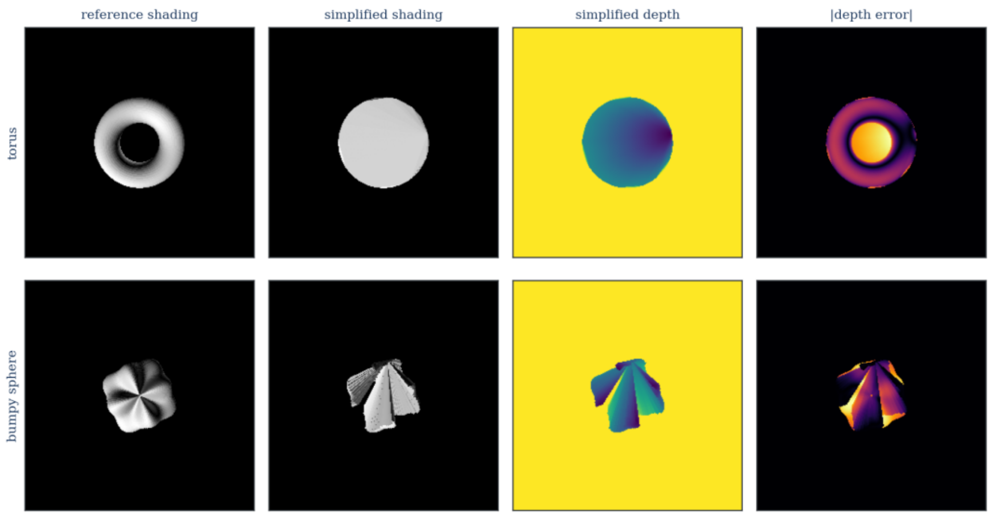

Figure 7: Qualitative comparison using the reproduction meshes. From left to right, the panels show reference shading, simplified shading, simplified depth, and absolute depth error. The remaining error is concentrated around silhouettes and depth discontinuities rather than across the surface interior, confirming that the dominant loss occurs in the depth buffer.

7.4 Failure analysis

The remaining score loss comes mainly from three sources, listed in approximate order of importance.

(i) Silhouette quantisation. After simplification, a curved silhouette is represented by a coarser polygonal boundary. Every SSIM window that crosses this boundary may be penalised, even when the interior surface is reconstructed accurately.

(ii) Thin features. Geometry that occupies only a few pixels in some views presents a difficult choice. It must either be retained at its full element cost or removed with a substantial score penalty. Discrete simplification offers little useful middle ground.

(iii) Occlusion ordering changes. When two surfaces are close in depth, a small change in vertex placement can determine which surface becomes visible at a pixel. A minor geometric displacement can therefore produce a much larger local depth error.

All three are primarily vertex placement problems rather than element selection problems. This is precisely the type of error that the current architecture cannot directly correct.

7.5 Successor design

A natural successor would replace the discrete evaluation gate with a differentiable soft rasteriser, following the methods in [6] and [7]. Probabilistic assignment between pixels and faces, controlled by a temperature parameter, would make the rendering process differentiable. SSIM could then be expressed as a differentiable function of the vertex positions, allowing those positions to be optimised directly using gradient descent while the topology remains fixed.

The proposed procedure would first use quadric collapse to reach the required element budget. The surviving vertex positions would then be optimised using a depth SSIM loss propagated through the soft rasteriser. During optimisation, the rasterisation temperature would gradually be reduced towards the hard rendering limit. This would encourage the final geometry to remain effective when evaluated using the true renderer.

This design directly addresses all three failure sources discussed in Section 7.4 because each of them depends on vertex placement. We estimated that implementing and validating this approach would require several days of additional work, which was beyond the available time for the round. We therefore present it as the most promising next direction rather than as a completed result.

7.6 Threats to validity

Three limitations should be considered when interpreting the results.

First, the channel decomposition was measured using our own implementation of the scorer. Although its aggregate output was validated against the official score, small differences in the shading model could change the absolute channel values. Such differences would be less likely to change the relative ordering that motivated our design.

Second, the controlled reproduction uses synthetic meshes and flat shading. It should therefore be viewed as directional evidence rather than an exact quantitative replication of the contest environment.

Third, the tier boundaries in Table 2 were tuned using the provided test set. They would likely need to be adjusted for a substantially different distribution of mesh sizes or geometric complexity.

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